



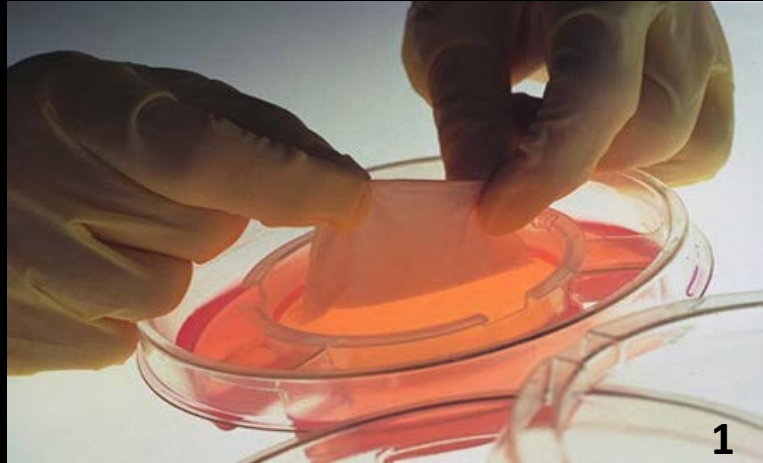
Department of Mechanical
Engineering University of California,
Berkeley

Design-nano.berkeley.edu

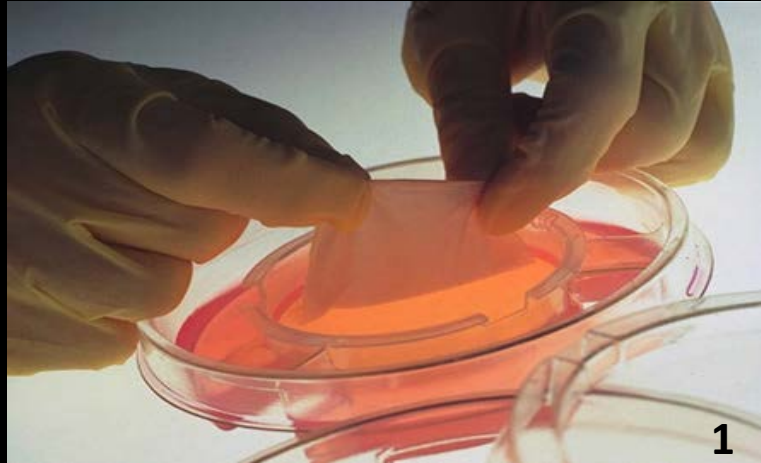
Micro-engineering
a platform
to
reconstruct physiology and
functionality
of
the human brain
microvasculature
in vitro

Yasaman Daghighi, Hossein Heidari,
Hayden Taylor

Tissue Engineering



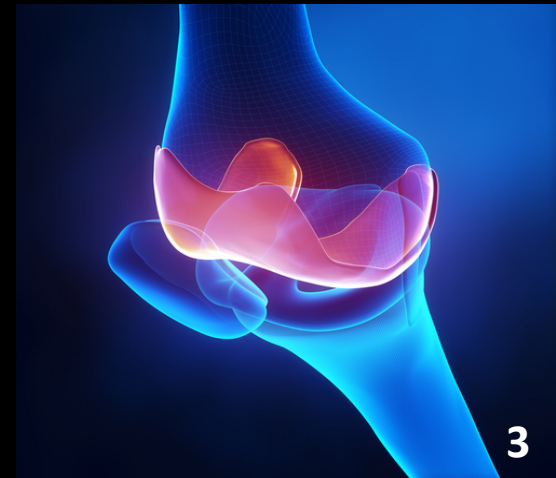
Successful Engineered Tissues



Burn Wound Healing Skin



Regenerate Cartilage

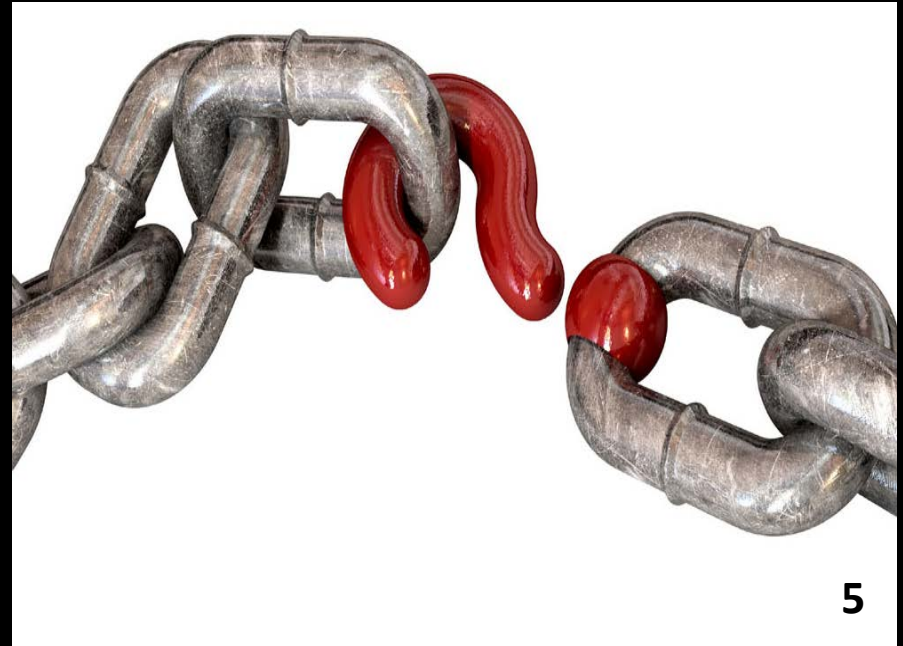
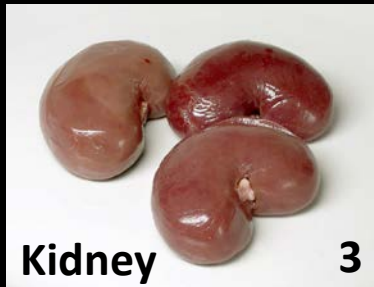
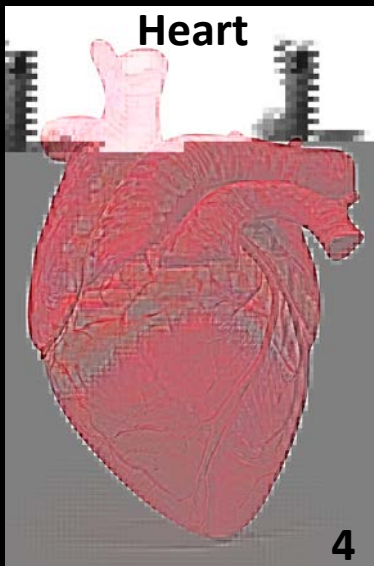
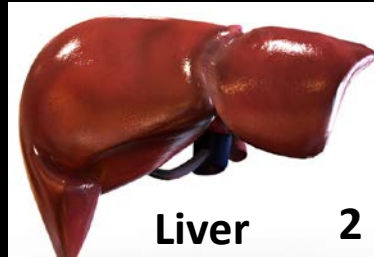
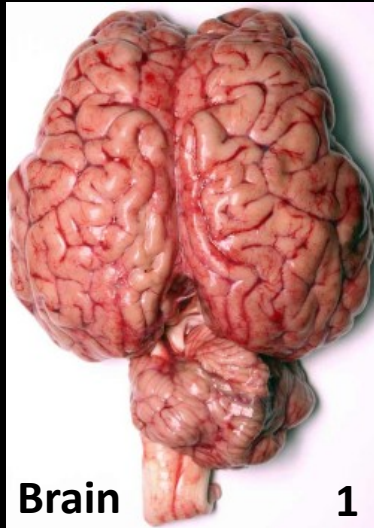


1- <https://prezi.com/e6hlori39pzy/tissue-engineering/>

2- <http://industryarcblog.com/applications-burn-wound-healing-drive-tissue-engineered-skin-substitute-market>

3- <https://www.biosciencetechnology.com/news/2014/05/engineers-grow-functional-human-cartilage-lab>

Microvasculature is the Missing Link in Evolution of Tissue Engineering Assays



1- <http://www.human-memory.net/brain.html>

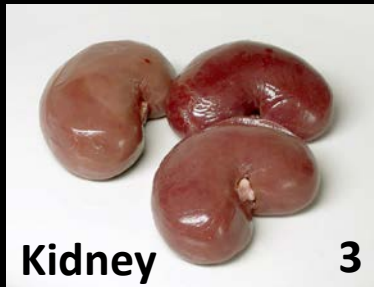
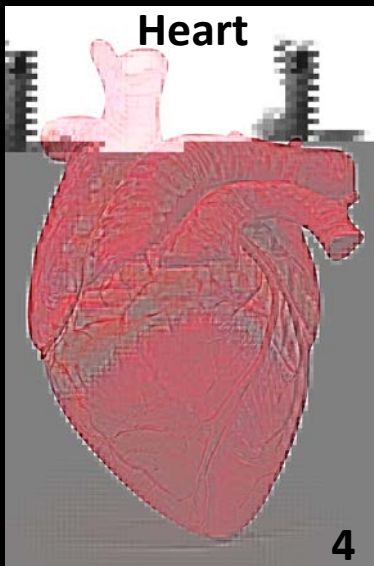
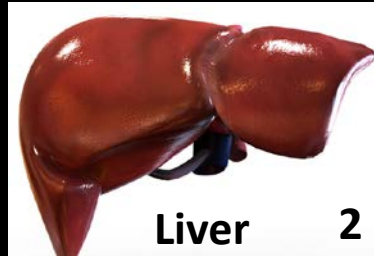
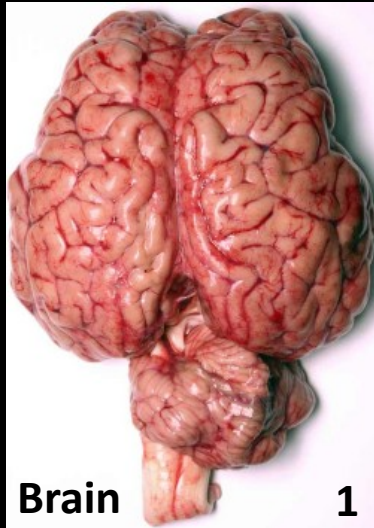
2- <https://www.highend3d.com/3d-model/human-liver-3d-model-83344>

3- <https://www.healthncure.org/functions-and-structure-of-kidney/>

4- <https://www.turbosquid.com/3d-models/human-heart-c4d/1008241>

5- <https://steamregister.com/myth-of-the-missing-link-in-evolution-does-science-no-favors-conv/>

Microvasculature is the Missing Link in Evolution of Tissue Engineering Assays



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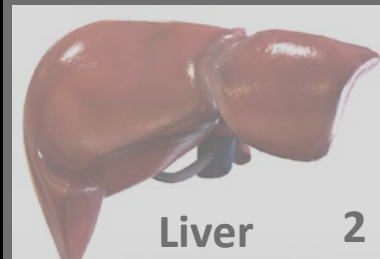
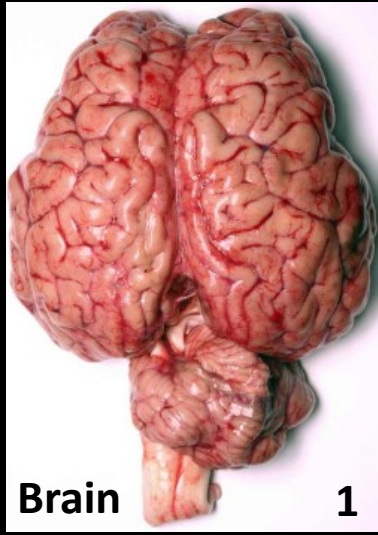
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Microvasculature is the Missing link in Evolution of Tissue Engineering Assays



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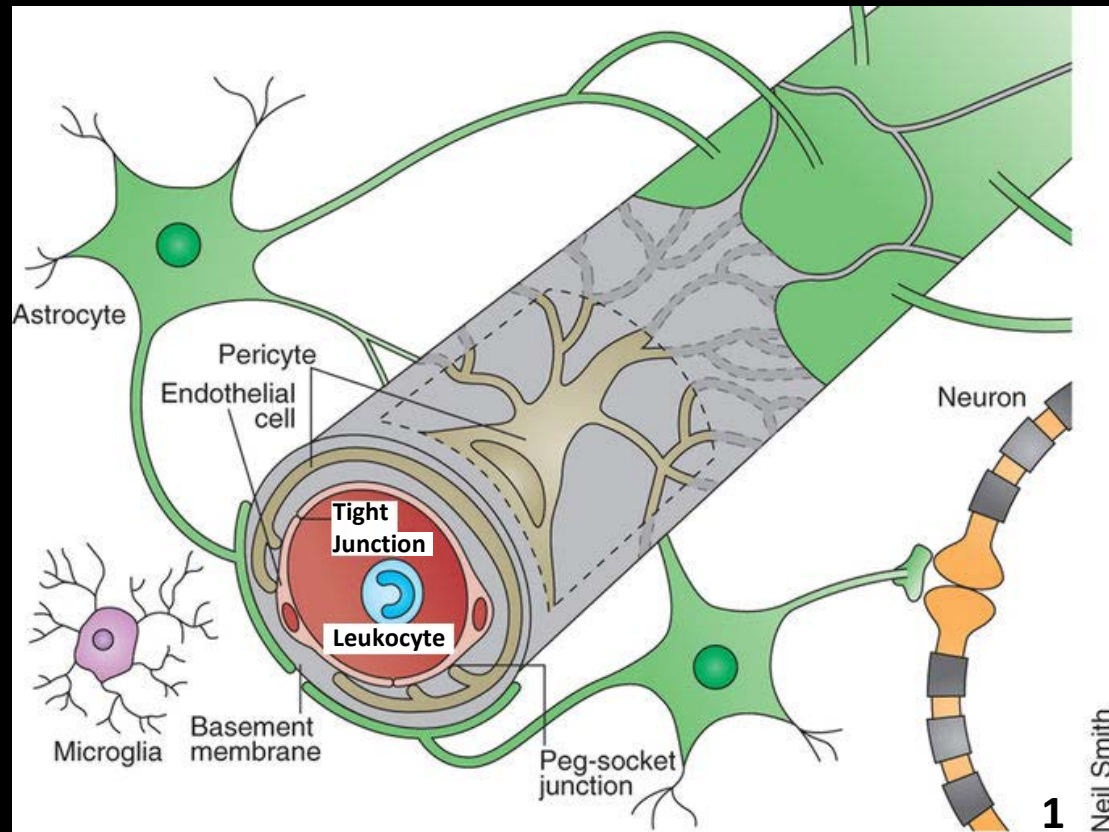
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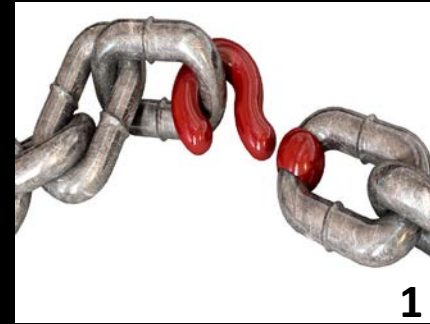
Layers Between Blood and Brain

- Endothelial cells covering the inner surface of the lumen
- Pericytes embedded in the basement membrane
- Astrocytes that surround the basement membrane



In Order to Address the Missing Link

- Co-culturing cell lines
 - **HMECs, Astrocytes, Pericytes**
- In a biologically-related environment
 - **Biocompatible ECMs**
- Using human cells
 - **Not animal's**
- Dimensions and shape of the lumen
 - **Covering wide range of cylindrical capillary sizes**
- Shear flow
- Transepithelial/transendothelial electrical resistance (TEER) test capability



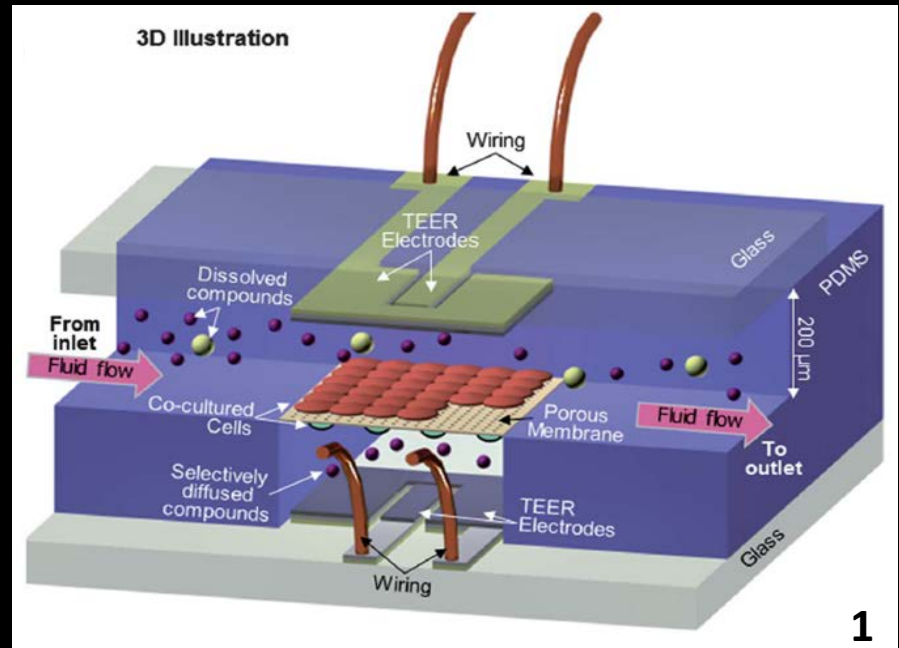
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Microfluidics-Based Strategies to Address the Missing Link

- ✗ • Co-culturing cell lines
- ✗ • In a biologically-related environment
- ✗ • Using human cells
- ✗ • Dimensions and **shape** of the lumen
- ✓ • Shear flow
- ✗ • TEER test capability

✗ **Used: membrane**

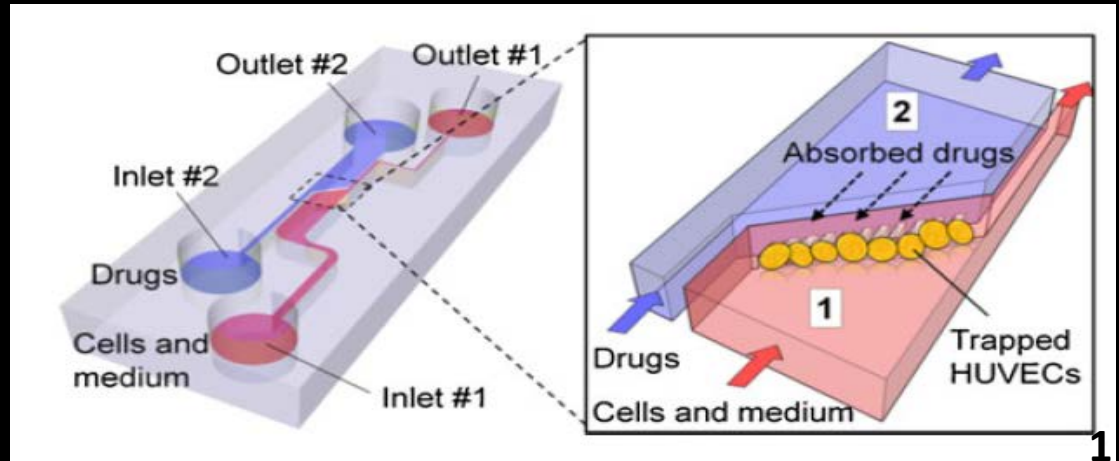


Microfluidics-Based Strategies to Address the Missing Link

- ✗ • Co-culturing cell lines
- ✗ • In a biologically-related environment
- ✗ • Using human cells
- ✗ • Dimensions and **shape** of the lumen
- ✓ • Shear flow
- ✗ • TEER test capability

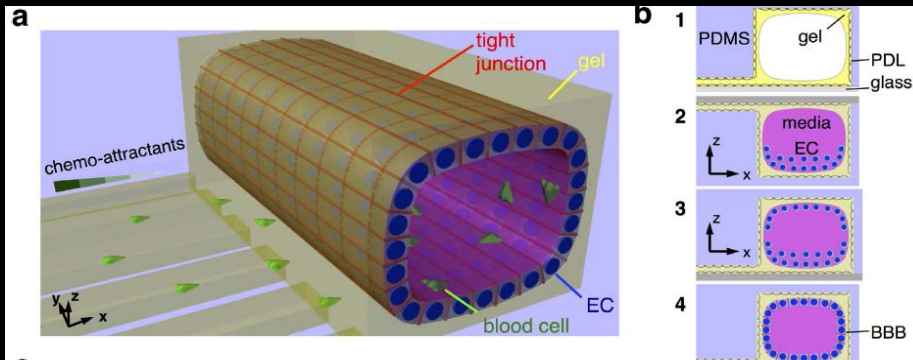
✗

Used: Holes on PDMS

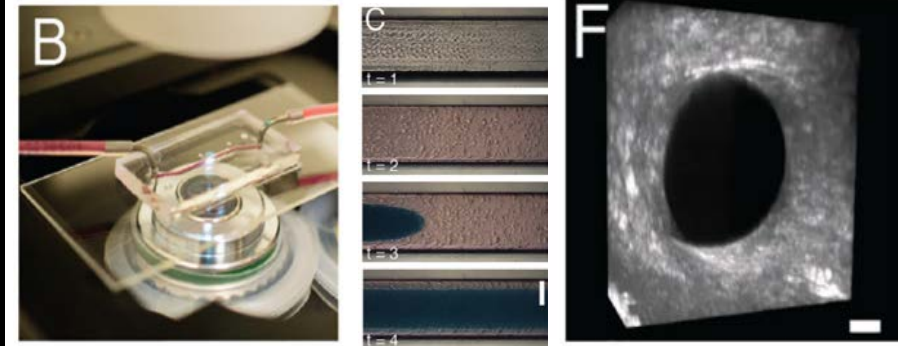


Microfluidics-Based Strategies to Address the Missing Link

- ✓✗ • Co-culturing cell lines
- ✓✗ • In a biologically-related environment
- ✓✗ • Using human cells
- ✗ • **Dimensions** and **shape** of the lumen
- ✓ • Shear flow
- ✓ • TEER test capability



Cho, H., Seo, J.H., Wong, K.H.K., Terasaki, et. al., and Irimia, D., Scientific Reports 5, 15222 (2015).



Herland, A., Van Der Meer, A.D., FitzGerald, E.A., Park, T.E., Sleeboom, J.J.F., and Ingber, D.E., PLoS ONE 11(3), 1–21 (2016).

Microfluidics-Based Strategies to Address the Missing Link

- Co-culturing cell lines
 - missing
- In a biologically-related environment
 - PDMS and glass provides a hard microenvironment for cells
 - membrane or holes on PDMS are not biologically-related
- Using human cells
 - missing
- Dimensions and shape of the lumen
 - rectangular channels, limited size increase, multi-layer microfabrication is complicated
- Shear flow
- TEER test capability



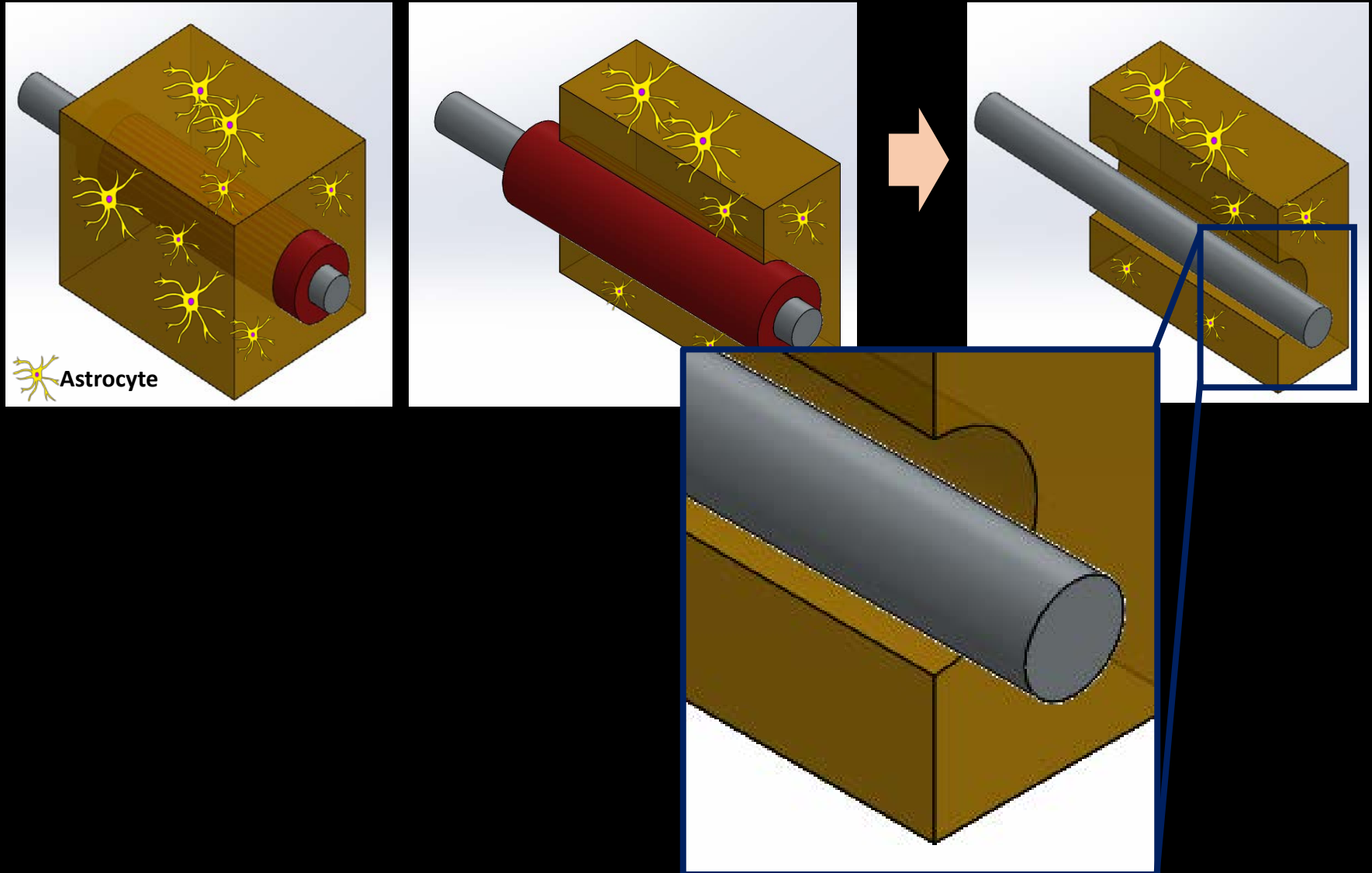
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Our Method Offers Solutions

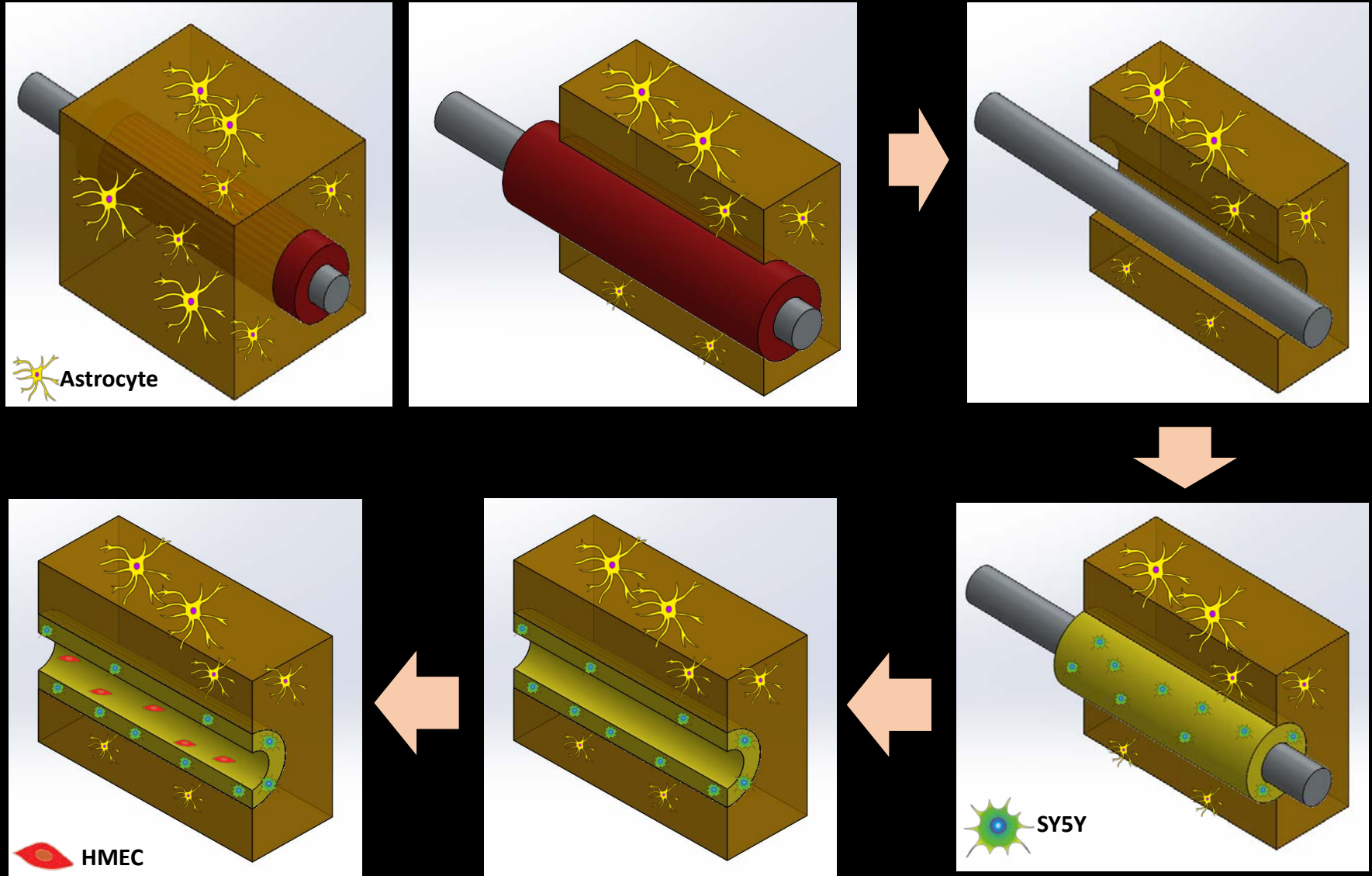
- Co-culturing cell lines
 - ✓ **HMECs, Astrocytes, SY5Ys**
- In a biologically-related environment
 - ✓ **No PDMS or glass used**
 - ✓ **No membrane or holes on PDMS used**
 - ✓ **Cells cultured in 3D assay in Hydrogel**
- Using human cells
 - ✓ **Used**
- Dimensions and shape of the lumen
 - ✓ **Cylindrical capillary is formed**
 - ✓ **Flexible in size range**
- **Shear flow** (future work)
- **TEER test capability** (future work)



Our Co-Axial 3D Culturing Method

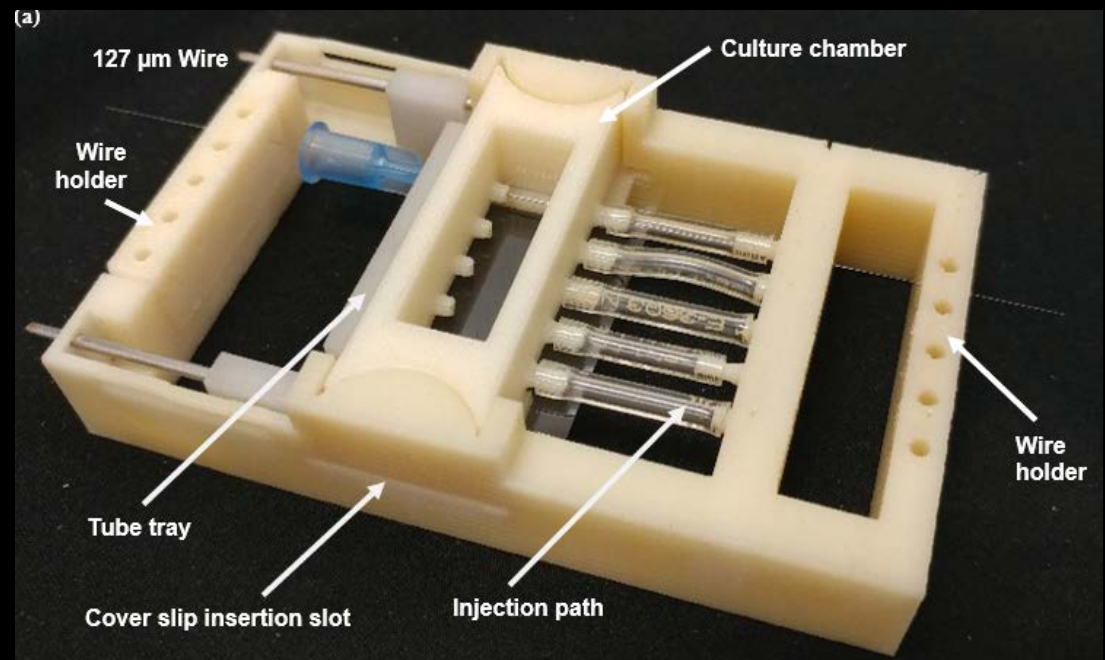


Our Co-Axial 3D Culturing Method

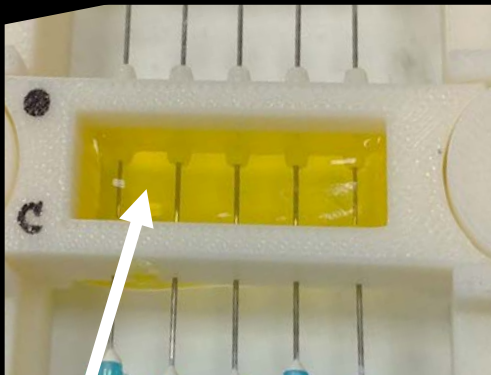
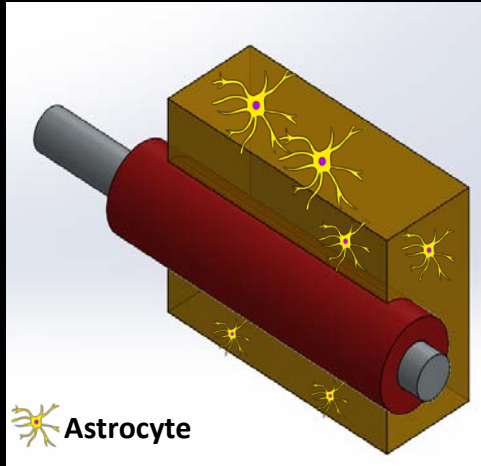


The beauty of using 3D printed platform

- Easy to use and rapid
- Costs less
- Customized complex design
- Mitigate Risk and receive feasible Feedback
- Manufacturing workflow
- Time-to-Market

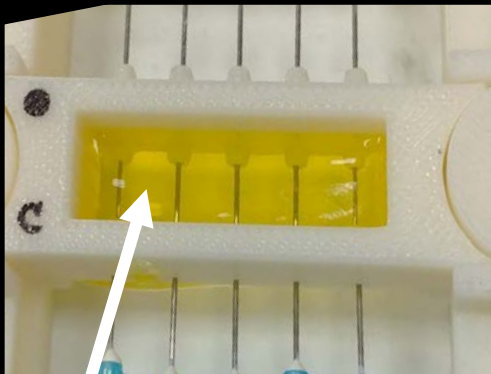
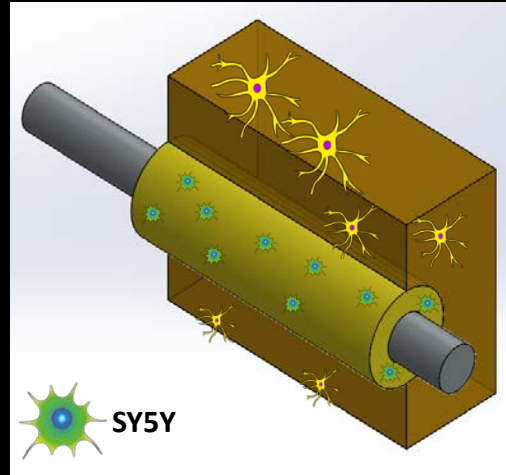
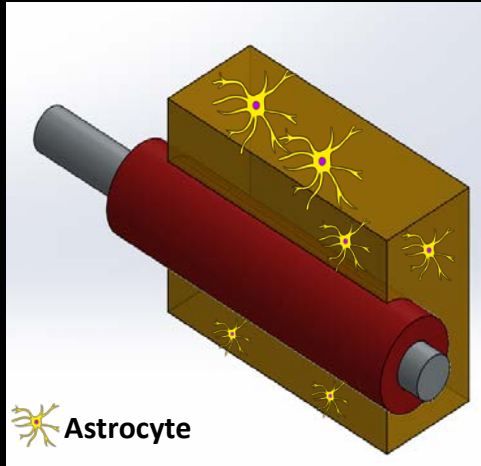


Testing the Prototype

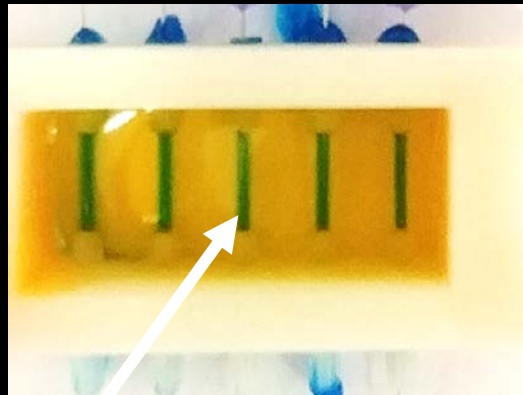


**Layer 1:
Agarose**

Testing the Prototype

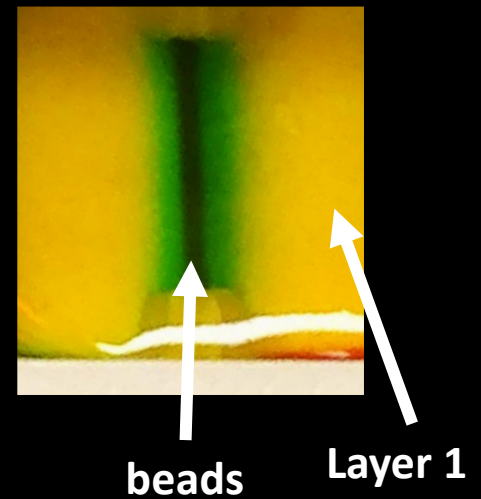
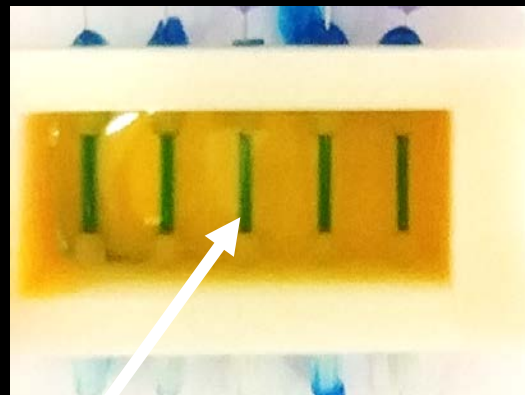
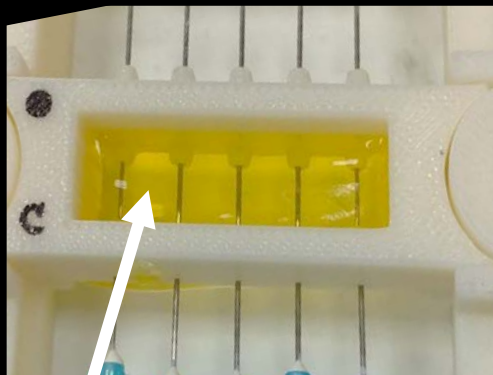
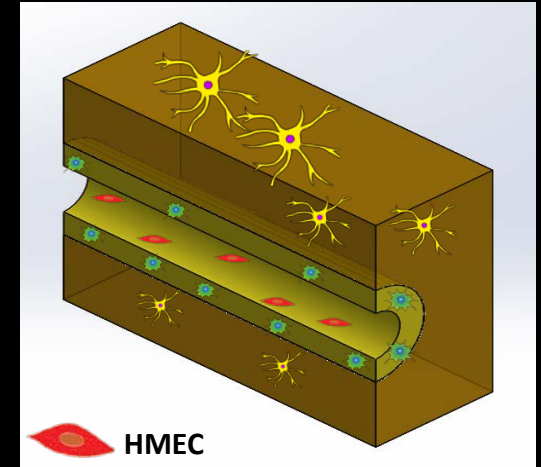
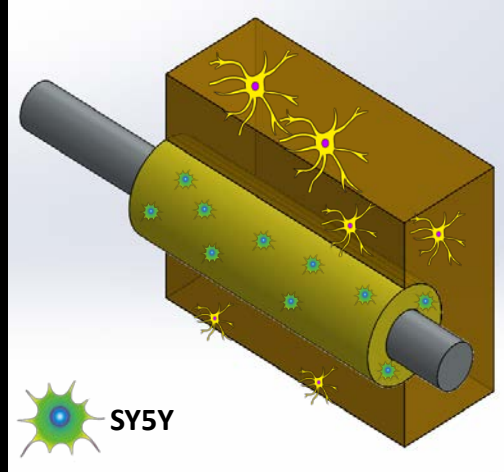
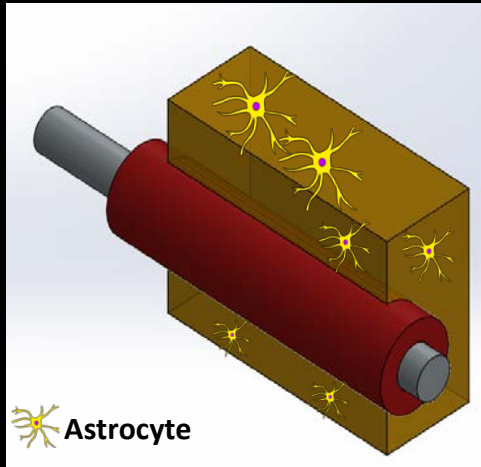


**Layer 1:
Agarose**

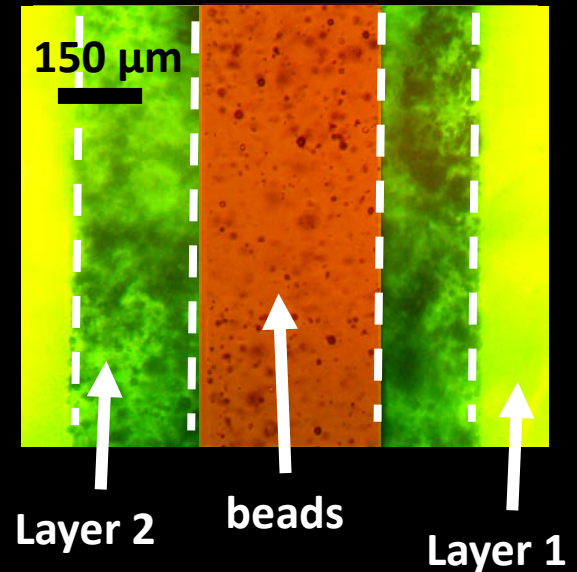
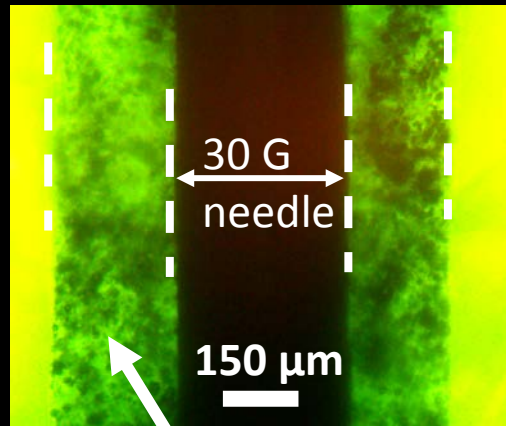
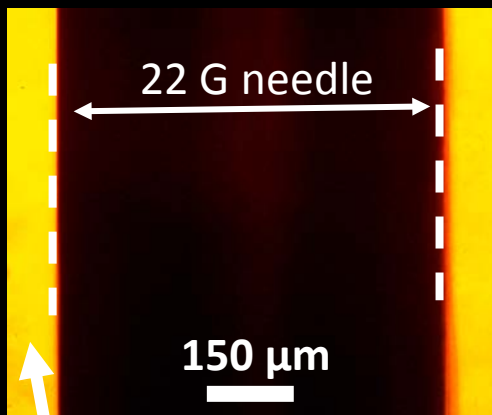
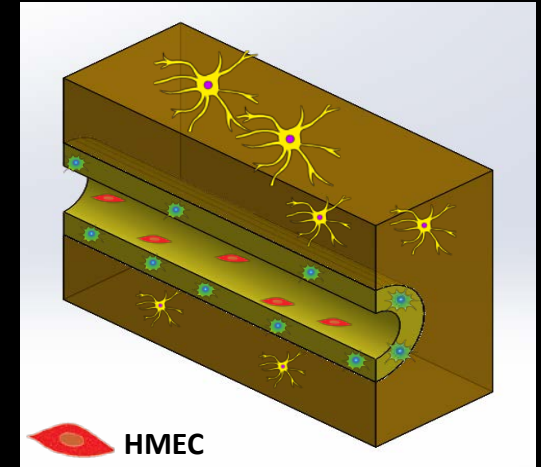
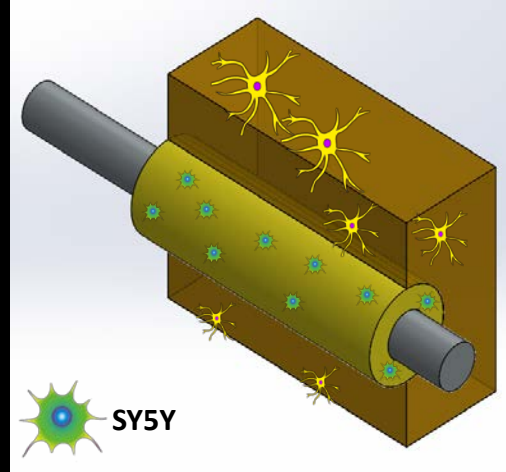
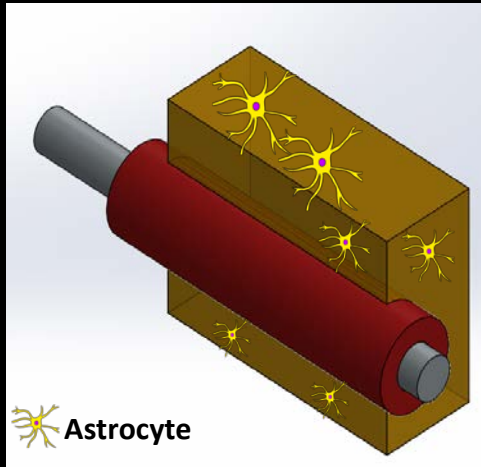


**Layer 2:
Agarose + beads**

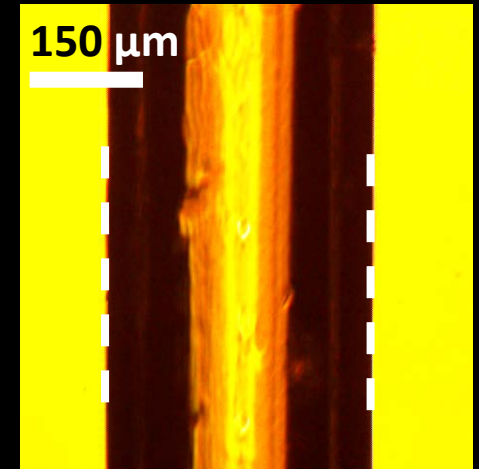
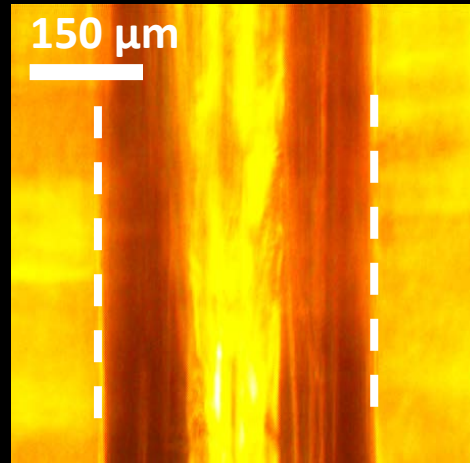
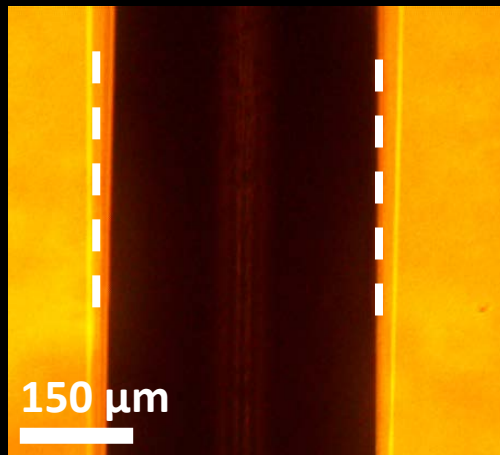
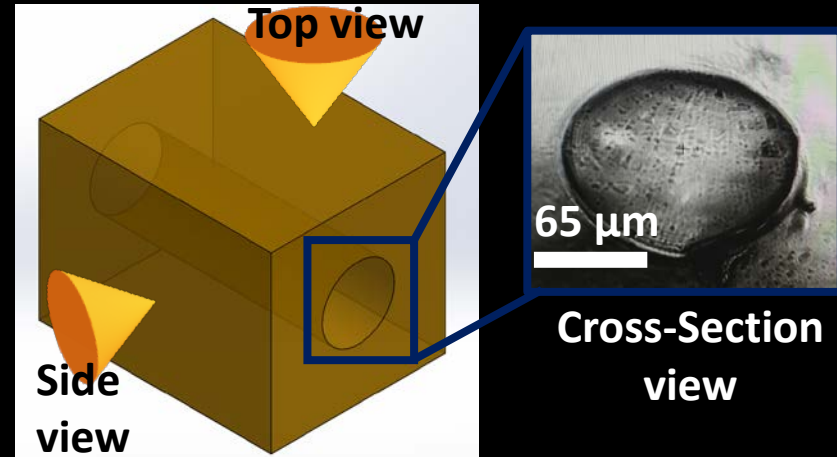
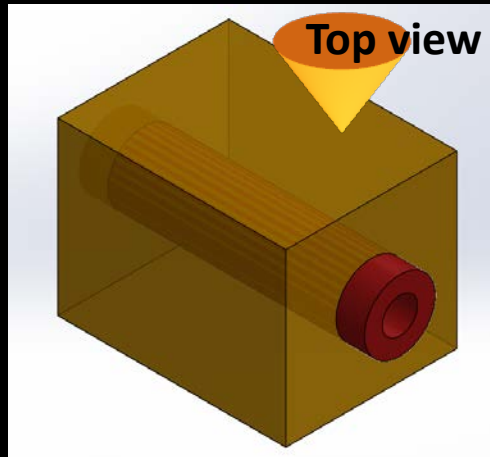
Testing the Prototype



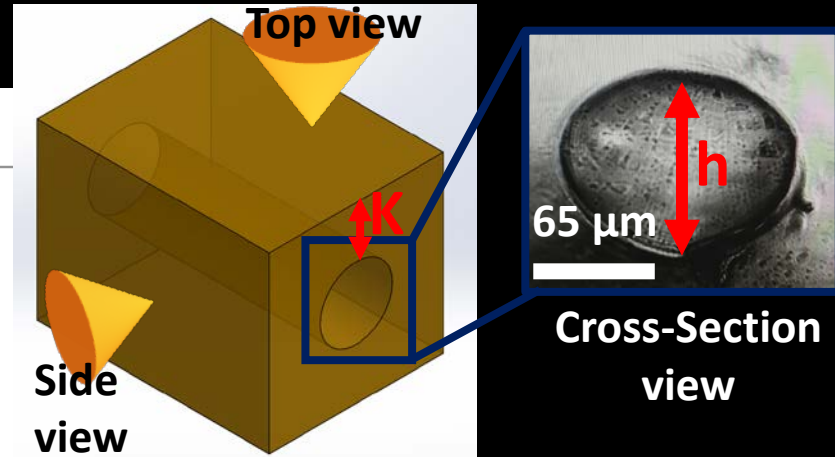
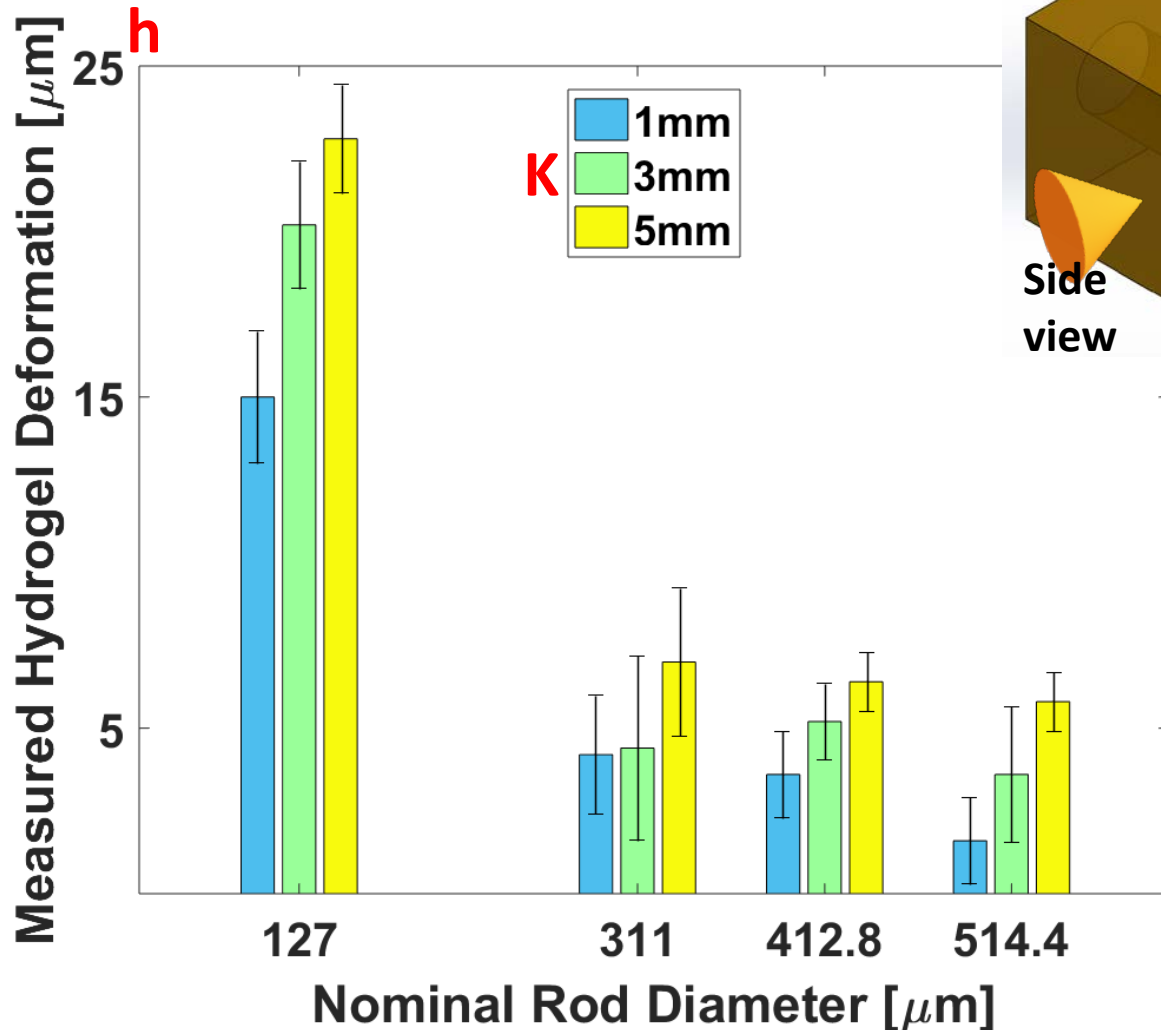
Testing the Prototype



Characterizing Gel Deformation in z Direction

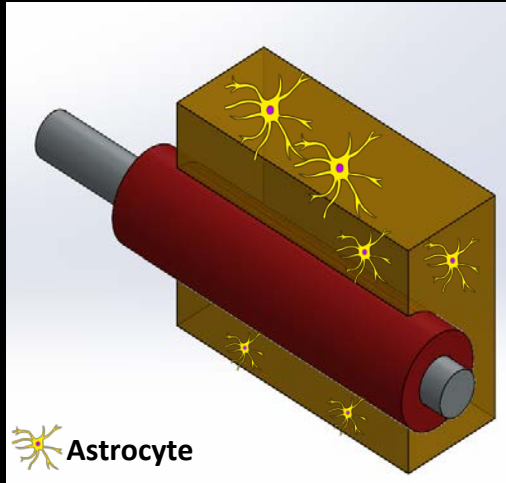


Characterizing Gel Deformation in z Direction

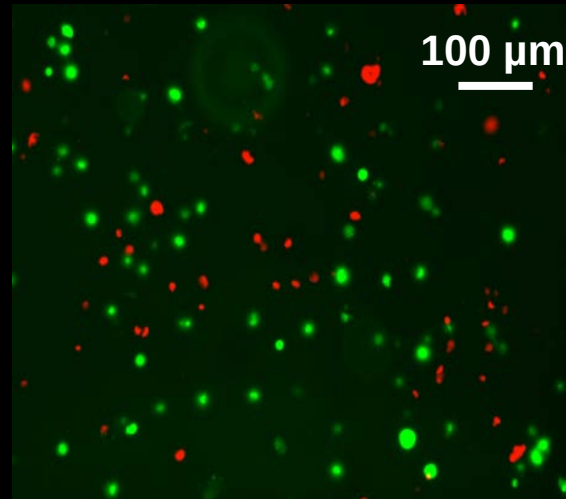


3D Culturing of Glial Cells

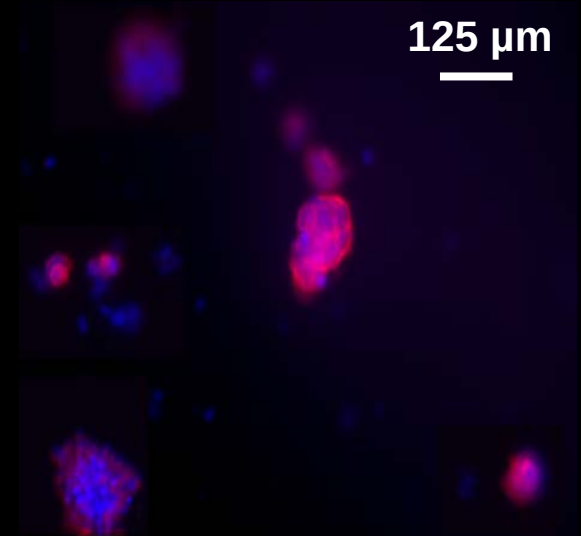
Astrocytes



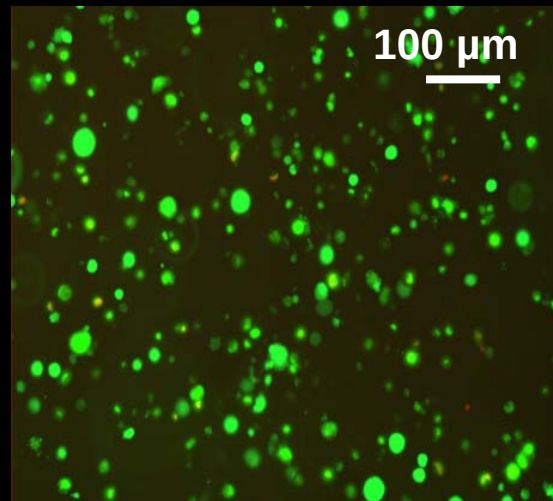
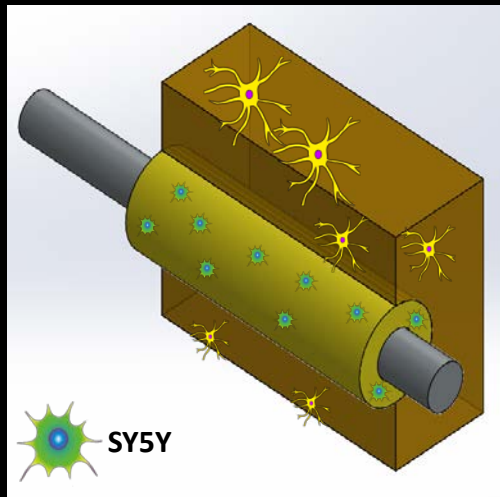
Live/dead assay



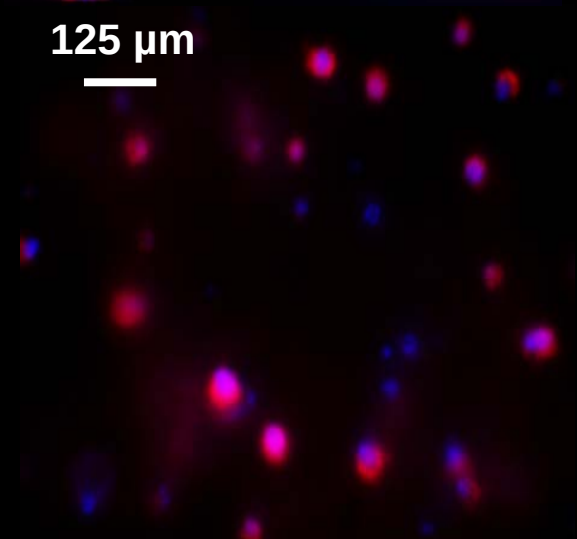
Proliferation



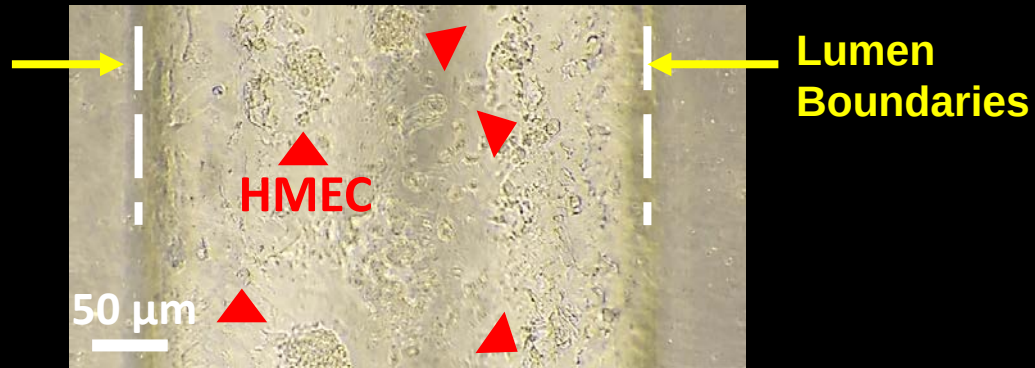
SY5Ys



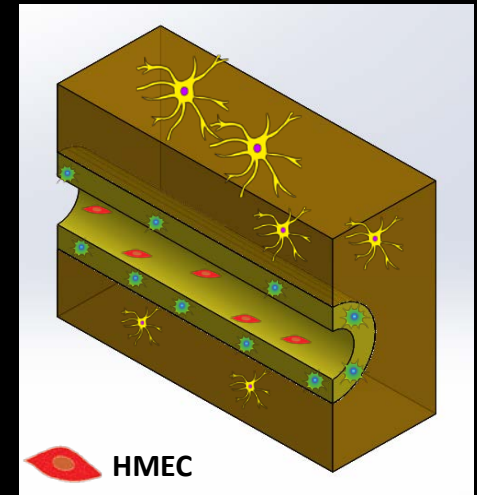
125 μm



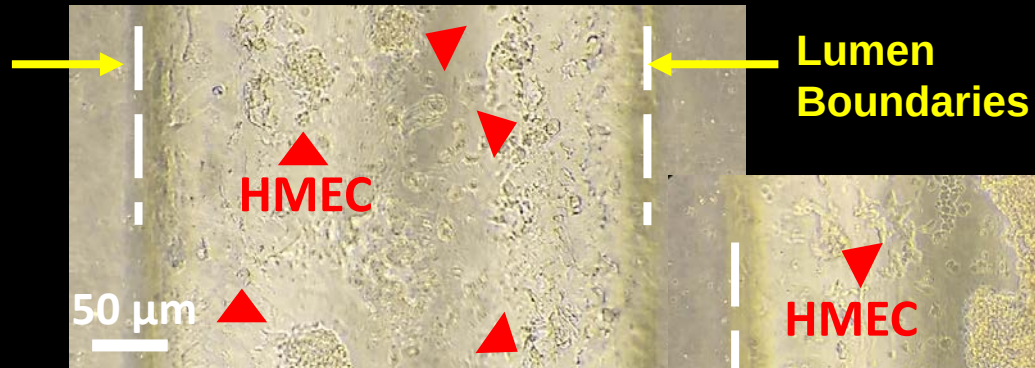
Culturing HMECs on Lumen Wall Inside Hydrogel



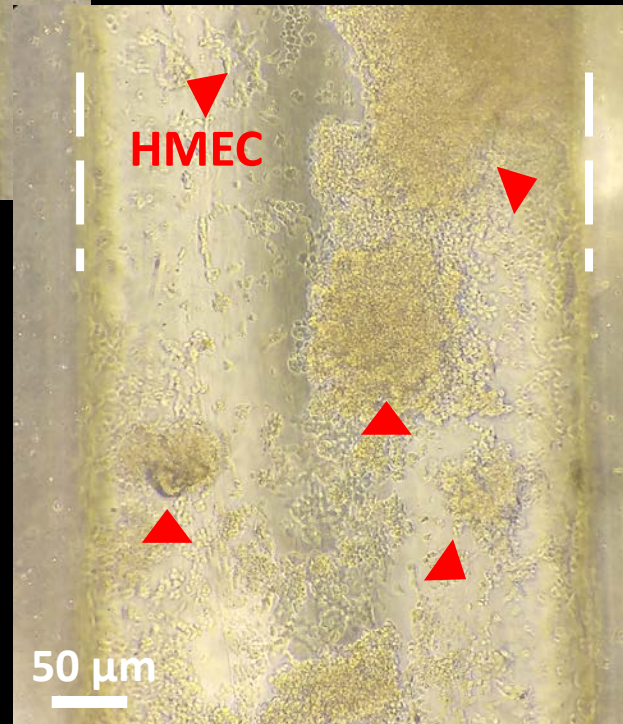
Day 2nd of culturing



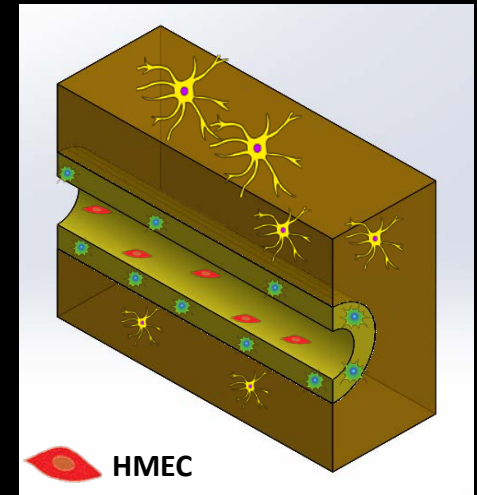
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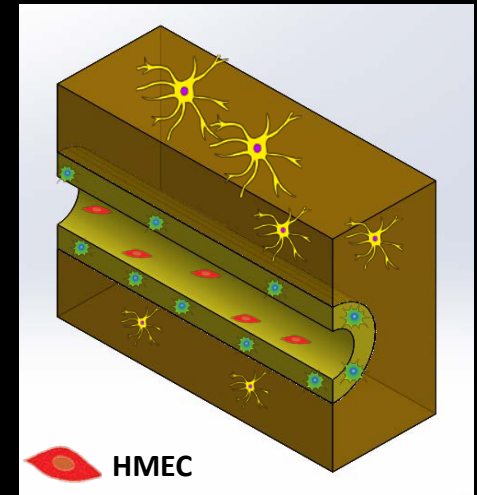
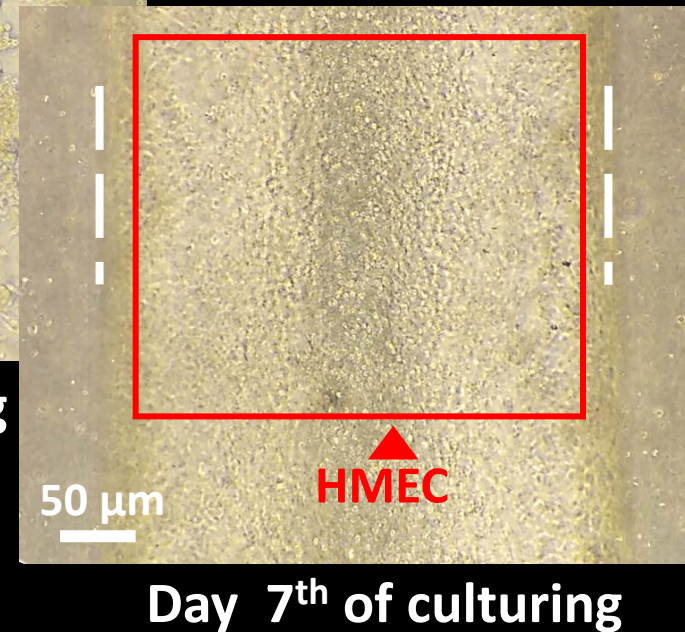
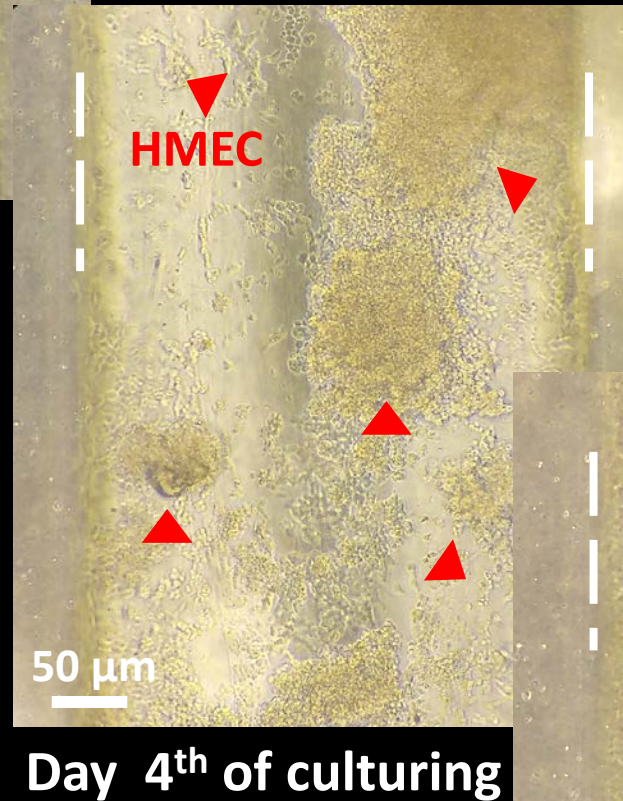
Day 2nd of culturing



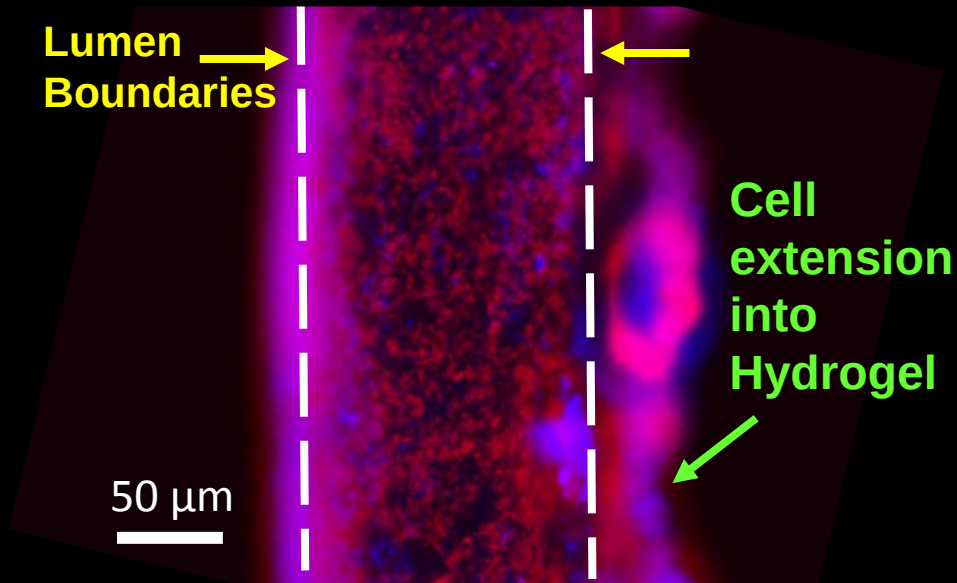
Day 4th of culturing



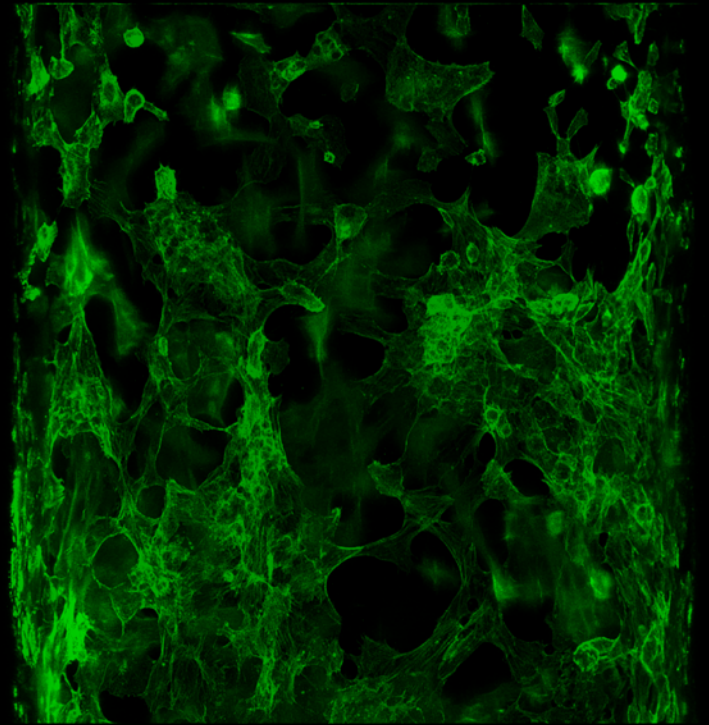
Culturing HMECs on Lumen Wall Inside Hydrogel



Engineered Microvasculature

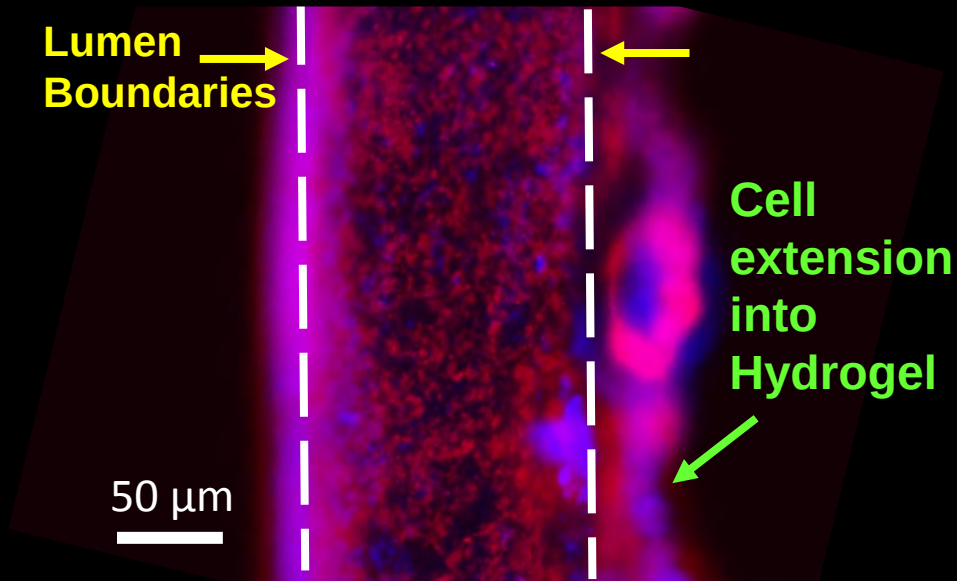


HMEC lumen
Day 7th of culturing

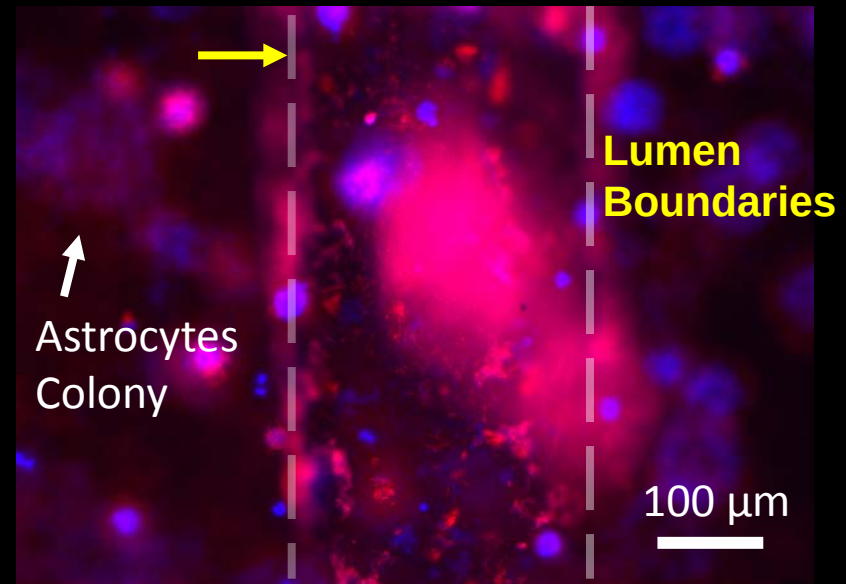


Confocal image of
HMEC lumen

Engineered Microvasculature



HMEC lumen
Day 7th of culturing



HMEC lumen and Astrocyte colonies
Day 7 of co-culturing

Achievements



- A 127 μm -diameter model brain vascular lumen has been successfully cultured *in vitro*
- In an agarose–collagen composite hydrogel
- using a new design of micromolding system by combining: 3D-printed casting framework + a coaxial stainless steel wire and needle
- Our system allows for the formation of multiple channels in parallel: five at present, with the potential for further expansion in future design iterations.
- This model could ultimately be used for **drug testing** and **investigating diseases** (in which the cerebral vasculature is relevant).

Thanks to

- **Our Lab members at Design-nano Berkeley**
- **Mr. B. Angelo and J. Lopez** (UC Berkeley Mechanical Engineering Student Machine Shop)
- **Dr. A. Killilea and Ms. C. Tasto** (UC Berkeley Cell Culture Facility)
- **Dr. Mary West** (UC Berkeley CIRM/QB3 Shared Stem Cell Facility and High-Throughput Screening Facility)
- UC Berkeley Mechanical Engineering faculty start-up funds.

Thank You



<https://www.utest.com/articles/question-whats-the-missing-link-in-the-qa-chain>